

D843

General Chemistry I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A lab at a steel mill that ships 400 tons a day analyzes the ion Fe^{3+} formed from iron-56 (atomic number 26). The mill's furnace runs at 1,500 °C. How many protons, neutrons, and electrons does the ion have?

- A. 26 protons, 30 neutrons, 23 electrons
- B. 26 protons, 56 neutrons, 23 electrons
- C. 26 protons, 30 neutrons, 26 electrons
- D. 26 protons, 30 neutrons, 29 electrons

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A fertilizer company that sells in 12 states is comparing four elements used in its products: Na, Mg, Cl, and K. Which has the largest atomic radius?

- A. Cl
- B. Mg
- C. K
- D. Na

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A cement plant that runs two kilns lines them with magnesium oxide (MgO). What type of bonding holds MgO together?

- A. Ionic bonding
- B. Covalent bonding, because the lining is heated in the kilns
- C. Polar covalent bonding
- D. Metallic bonding

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A beverage lab that bottles 5,000 cases a day weighs out 36.0 g of water (molar mass 18.0 g/mol). How many water molecules is that?

- A. 2.00
- B. 2.17×10^{25}
- C. 3.01×10^{23}
- D. 1.20×10^{24}

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, a fuel cell test at a lab with 14 researchers starts with 4.0 mol H_2 and 3.0 mol O_2 . What is the maximum amount of water that can form?

- A. 6.0 mol
- B. 7.0 mol
- C. 4.0 mol
- D. 3.0 mol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, how many moles of O_2 are needed to react completely with 5.0 mol H_2 ?

- A. 10.0 mol
- B. 2.5 mol
- C. 2.0 mol
- D. 5.0 mol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hospital pharmacy with nine technicians needs 500 mL of 0.30 M saline and has a 1.5 M stock solution. What volume of stock is needed?

- A. 400 mL
- B. 150 mL
- C. 100 mL
- D. 2,500 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

25.0 mL of 2.0 M HCl is diluted to a final volume of 250.0 mL. What is the new concentration?

- A. 2.0 M, since adding water never changes molarity
- B. 20 M
- C. 0.18 M
- D. 0.20 M

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A brewery in a city of 200,000 tests two batches. Batch A has pH 3.0 and batch B has pH 5.0. How many times greater is $[H^+]$ in batch A than in batch B?

- A. 100 times
- B. 0.01 times
- C. About 1.67 times
- D. 2 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A solution has $[H^+] = 1 \times 10^{-4}$ M. What is its pH?

- A. 0.0001
- B. 4
- C. -4
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hospital nuclear medicine department that opened in 2015 receives 80 mg of an isotope with a half-life of 5 days. How much remains after 15 days?

- A. 0 mg, since it decays completely
- B. 40 mg
- C. About 26.7 mg
- D. 10 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A 64 g radioactive sample decays to 4 g in 20 hours. What is its half-life?

A. 1.25 hours

B. 5 hours

C. 20 hours

D. It cannot be found, since half-life always depends on starting mass.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A lab at a steel mill ~~that ships 400 tons a day~~ analyzes the ion Fe^{3+} formed from iron-56 (atomic number 26). ~~The mill's furnace runs at 1,500 °C.~~ How many protons, neutrons, and electrons does the ion have?

Target word: particle counts in an ion **Decides:** $Z = 26$, $A = 56$, charge 3+.

A	Correct	$56 - 26 = 30$ neutrons; a 3+ ion has lost 3 electrons.
B	Wrong stem number	Uses the mass number as the neutron count.
C	Dropped element	Ignores the 3+ charge.
D	Inverted operation	Adds electrons; a positive ion has lost them.

2. Answer C Concept 2

A fertilizer company ~~that sells in 12 states~~ is comparing four elements used in its products: Na, Mg, Cl, and K. Which has the largest atomic radius?

Target word: largest atom **Decides:** K is one row below Na in the same group, at the far left.

A	Inverted operation	Radius shrinks across a period; Cl is the smallest here.
B	Confusion set	Mistakes a slightly higher atomic mass for a bigger atom; Mg is smaller than Na.
C	Correct	Down a group adds an electron shell; K is below Na.
D	Dropped element	Applies the across-period trend but drops the down-group trend.

3. Answer A Concept 3

A cement plant ~~that runs two kilns~~ lines them with magnesium oxide (MgO). What type of bonding holds MgO together?

Target word: bond type **Decides:** Mg is a metal; O is a nonmetal.

A	Correct	Mg transfers two electrons to O, forming Mg^{2+} and O^{2-} .
B	Camouflage	Heating does not decide bond type.
C	Adjacent category	Sharing between nonmetals with unequal pull; MgO involves transfer.
D	Confusion set	Occurs among metal atoms only.

4. Answer D Concept 4

A beverage lab ~~that bottles 5,000 cases a day~~ weighs out 36.0 g of water (molar mass 18.0 g/mol). How many water molecules is that?

Target word: number of molecules **Decides:** $36.0 \div 18.0 = 2.00$ mol.

A	Dropped element	That is moles, not molecules.
B	Dropped element	$36.0 \times 6.022 \times 10^{23}$ skips dividing by molar mass.
C	Inverted operation	Divides Avogadro's number by 2 instead of multiplying.
D	Correct	$2.00 \text{ mol} \times 6.022 \times 10^{23} = 1.20 \times 10^{24}$.

5. Answer C Concept 5

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, a fuel cell test at a lab with 14 researchers starts with 4.0 mol H_2 and 3.0 mol O_2 . What is the maximum amount of water that can form?

Target word: maximum product **Decides:** 4.0 mol H_2 needs only 2.0 mol O_2 , so H_2 is limiting.

A	Inverted operation	Treats O_2 as limiting (3.0×2); there is not enough H_2 .
B	Dropped element	Adds reactant moles and drops the mole ratios.
C	Correct	$4.0 \text{ mol } \text{H}_2 \times (2 \text{ H}_2\text{O} \div 2 \text{ H}_2) = 4.0 \text{ mol H}_2\text{O}$.
D	Wrong stem number	The O_2 amount.

6. Answer B Concept 5 • transfer

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, how many moles of O_2 are needed to react completely with 5.0 mol H_2 ?

Target word: solve for the other reactant **Decides:** Ratio 1 O_2 : 2 H_2 .

A	Inverted operation	Multiplies by 2 instead of dividing.
B	Correct	$5.0 \times (1 \div 2) = 2.5$.
C	Wrong stem number	A coefficient from the equation.
D	Dropped element	Ignores the 2:1 ratio.

7. Answer C Concept 6

A hospital pharmacy with nine technicians needs 500 mL of 0.30 M saline and has a 1.5 M stock solution. What volume of stock is needed?

Target word: volume of stock **Decides:** $V_1 = (0.30 \times 500) \div 1.5$.

A	Adjacent category	That is the water to add, not the stock.
B	Dropped element	0.30×500 without dividing by the stock concentration.
C	Correct	$150 \div 1.5 = 100 \text{ mL}$, then dilute to 500 mL.
D	Inverted operation	Multiplies by $1.5 \div 0.30$ instead of $0.30 \div 1.5$.

8. Answer D Concept 6 • transfer

25.0 mL of 2.0 M HCl is diluted to a final volume of 250.0 mL. What is the new concentration?

Target word: solve for M_2 **Decides:** $M_2 = 2.0 \times 25.0 \div 250.0$.

A	Absolute language	Dilution lowers concentration.
B	Inverted operation	Multiplies by $250 \div 25$ instead of dividing.
C	Adjacent category	Treats 250 mL as water added (275 mL total), not the final volume.
D	Correct	$50 \div 250 = 0.20 \text{ M}$.

9. Answer A Concept 7

A brewery in a city of 200,000 tests two batches. Batch A has pH 3.0 and batch B has pH 5.0. How many times greater is $[H^+]$ in batch A than in batch B?

Target word: ratio of $[H^+]$ **Decides:** Two pH units = 10^2 .

A	Correct	$10^{-3} \div 10^{-5} = 100$.
B	Inverted operation	Compares B to A.
C	Confusion set	Divides the pH values ($5 \div 3$) instead of comparing $[H^+]$.
D	Dropped element	Reads the pH difference as linear.

10. Answer B Concept 7 • transfer

A solution has $[H^+] = 1 \times 10^{-4}$ M. What is its pH?

Target word: solve for pH **Decides:** $pH = -\log(10^{-4})$.

A	Wrong stem number	The concentration itself.
B	Correct	$-\log(10^{-4}) = 4$.
C	Inverted operation	Drops the negative sign in the definition.
D	Confusion set	That is the pOH ($14 - 4$).

11. Answer D Concept 8

A hospital nuclear medicine department that opened in 2015 receives 80 mg of an isotope with a half-life of 5 days. How much remains after 15 days?

Target word: amount remaining **Decides:** 15 days = 3 half-lives.

A	Absolute language	Each half-life removes half of what remains.
B	Dropped element	Applies only one half-life.
C	Inverted operation	Divides by 3 instead of halving three times.
D	Correct	$80 \rightarrow 40 \rightarrow 20 \rightarrow 10$.

12. Answer B Concept 8 • transfer

A 64 g radioactive sample decays to 4 g in 20 hours. What is its half-life?

Target word: solve for half-life **Decides:** $64 \rightarrow 4$ is a factor of $16 = 2^4$, so 4 half-lives.

A	Inverted operation	Divides by 16 instead of by 4 half-lives.
B	Correct	$20 \div 4 = 5$.
C	Wrong stem number	The total elapsed time.
D	Absolute language	Half-life is independent of sample size.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Protons, neutrons, and electrons	Protons = atomic number; neutrons = mass number – atomic number; a positive ion has lost electrons equal to its charge.	Using the mass number as the neutron count or ignoring the ion's charge.	An isotope name or mass number plus an ion charge.
2	Periodic trends in atomic radius	Atomic radius shrinks left to right across a period and grows top to bottom down a group.	Assuming heavier or higher-numbered atoms are always bigger across a period.	Elements from different rows and columns compared by size.
3	Ionic vs. covalent bonding	A metal and a nonmetal usually form an ionic bond by transferring electrons; nonmetals share electrons in covalent bonds.	Calling any compound of two elements covalent.	Whether the formula pairs a metal with a nonmetal.
4	The mole	Moles = mass ÷ molar mass; particles = moles × 6.022×10^{23} .	Stopping at moles or skipping the molar mass step.	A mass in grams and a question about number of particles.
5	Stoichiometry and limiting reactant	Use mole ratios from the balanced equation; the reactant that runs out first sets the maximum product.	Using the reactant with fewer moles or ignoring coefficients.	Two reactant amounts and a “maximum product” question.
6	Molarity and dilution	$M_1V_1 = M_2V_2$, where V_2 is the final total volume.	Inverting the ratio or treating the added water as the final volume.	A stock concentration and a target concentration and volume.
7	pH is logarithmic	$\text{pH} = -\log[\text{H}^+]$; each one-unit drop in pH means ten times more H^+ .	Treating pH differences as linear.	Two pH values compared, or an $[\text{H}^+]$ in scientific notation.
8	Half-life	Remaining amount = initial × $(1/2)^{(\text{time} \div \text{half-life})}$.	Subtracting or dividing linearly instead of halving repeatedly.	An initial amount, a half-life, and an elapsed time.

Course scope (public description): General Chemistry I introduces foundational principles of chemistry, starting at the atomic level and expanding to the behavior of elements within the periodic table. This course explores how atoms bond to form molecules and proceeds into chemical reactions, acids and bases, solutions, and nuclear reactions. Students will gain a comprehensive understanding of stability and change in chemical processes. This course highlights the practical aspects of chemistry, providing insights into how chemical principles underpin everyday phenomena and contribute to our understanding of environmental processes.